
Model Provenance via Shard Membership Auditing

Anonymous Author(s)

Abstract

1 Foundational models are increasingly adapted for downstream applications, pro-
2 ducing large number of derivative models. This motivates the *model provenance*
3 problem: Given a target model and a candidate parent model, can one test, with
4 statistical guarantees, whether the target was derived from the parent? Reliable
5 answers are essential for intellectual property disputes, vulnerability tracing, and
6 accountability, yet existing tests rest on evidence that is not causally motivated.
7 We address this gap by making the lineage relation explicit, where we carefully
8 analyze the underlying assumptions. We propose a general statistical framework
9 for black-box model provenance that reduces the problem to data membership au-
10 diting. We test whether the target model inherits knowledge of a secret shard such
11 as a subset of training data or licensed corpus from the parent candidate model.
12 This reduction yields provenance tests with rigorous statistical guarantees. We in-
13 stantiate our framework for LLM provenance, testing whether a target LLM was
14 fine-tuned from a candidate parent, and demonstrate its effectiveness across a di-
15 verse set of models and datasets.

16 1 Introduction

17 Foundation models have become reusable building blocks for modern machine-learning systems.
18 Rather than training models from scratch, practitioners increasingly start from public checkpoints,
19 proprietary base models or open-source model families, and then adapt them through methods such
20 as fine-tuning, distillation or model merging. This reuse has motivated the *model provenance* prob-
21 lem: Given a target model and a candidate parent, can we determine whether the target was derived
22 from the parent? This lineage relation matters for intellectual-property disputes, license compliance,
23 vulnerability tracing, data-governance obligations, scientific accountability [Tramèr et al., 2016,
24 Orekondy et al., 2019, Carlini et al., 2024, Longpre et al., 2024].

25 One challenge in reliable model provenance is that the derivation process is often indirect and un-
26 observed, i.e., it can produce overlapping behavioral signatures, while the relevant training data
27 records and model internals are often unavailable. This is especially important in black-box auditing
28 settings, where an external party may need to assess provenance without access to model weights
29 or the training pipeline. Recent work shows that model provenance can nevertheless be approached
30 statistically with only black-box access to the models [Nikolic et al., 2025, Kuditipudi et al., 2025].
31 Nikolic et al. [2025] propose a black-box test for LLMs that compares the output similarity between
32 a target model and a candidate parent against the similarities between the target and a set of control
33 models. Kuditipudi et al. [2025] formulate black-box provenance through palimpsestic member-
34 ship inference, testing whether a target model is statistically dependent on randomized traces of the
35 parent training run, such as the order in which training examples were observed.

36 However, the observable evidence used to support a lineage claim is not always the information
37 whose transmission makes provenance important. In Nikolic et al. [2025], the evidence is relative
38 output similarity, whose interpretation depends on whether the chosen control models accurately
39 represent the no-lineage null. In Kuditipudi et al. [2025], the evidence comes from dependence on
40 the order in which training examples were observed; this provides strong evidence of association
41 when the training order is available, retained, and randomized independently of other confounding

factors, but the order itself is usually not the inheritance one ultimately wants to establish. In many provenance disputes, the central question is instead whether the target inherited specific information carried by the candidate parent, such as a private shard, curated corpus, licensed subset, or domain-specific training signal.

This motivates a formulation in which the object being tested is aligned with the reason provenance matters. Thus, it raises a natural question:

Can black-box model provenance be statistically tested through the information whose transmission makes provenance important.

We approach this question by reducing black-box provenance to a *data membership auditing* problem. The tester identifies a secret shard S whose transmission would be provenance-relevant, such as a private subset or a licensed corpus. The goal is to test whether the target model T exhibits statistical dependence on S . Under the causal assumption that T could not have accessed S except through the candidate parent P , such dependence supports a provenance claim from P to T .

Contributions.

2 Related Work

Model provenance. We focus on black-box model provenance, noting a number of difference to what is accessible to the auditor. While Nikolic et al. [2025] assumes no access to both the parent and the target model, it relies on a set of control models that are trained independently of the parent, whose validity is crucial to the underlying hypothesis test. The statistical soundness thus rely on this control set being sufficiently large and representative of independently trained models, which may be difficult to verify in practice. In contrast, Kuditiipudi et al. [2025] assumes knowledge of the training data ordering, but only black-box access to the target model. Their approach relies on the ability to randomize the training process of the parent model, which may not be feasible in many real-world scenarios where the parent model is already trained and deployed. The statistical guarantees of their method depend on the ability to effectively randomize the training process and the assumption that the target model’s behavior is influenced by the training data ordering. Our work differs from these approaches by focusing on testing certain information, such as a private shard or a licensed corpus part of the parent model’s training data, and by providing an estimator bounding both Type-I and Type-II errors. This allows us to provide statistical guarantees on the test results without relying on assumptions about control models or training data randomization. Maini et al. [2021] propose a white-box test framed as a dataset inference problem, using prediction-margin statistics, and hence requiring white-box access, on the owner’s private training data to test whether a suspect classifier is trained on that data. Our framework focuses on testing the derivation from a parent model, where our null is stated at the joint-distribution level. Our work provides general **Teodora: distribution-free?** results on the test’s Type-I and Type-II errors, while Maini et al. [2021] only provide guarantees on the Type-I error.

Model fingerprinting. Lukas et al. [2019] [Xu et al., 2024] [Pasquini et al., 2025]

Auditing memorization.

3 Preliminaries

Teodora: do we need preliminaries?

4 A Statistical Framework for Model Provenance

We consider an audit setting in which the tester knows a subset of training data S of the candidate parent P , which we call a *secret shard*. We assume the shard S is private, even though the underlying population $\mathcal{D}$ from which they are drawn may be public or partially known. Formally, Let $\mathcal{X}$ be some domain and let $\mathcal{D}$ be a distribution on $\mathcal{X}$. A *secret shard* is a random variable

$$S = (x_1, \dots, x_n) \sim \mathcal{D}^n.$$

Our framework reduces model provenance to a membership-auditing-style question: can the tester detect statistical dependence between the target model T and the secret shard S ? A "Yes" answer establishes only a statistical claim: T appears to depend on S . To interpret this dependence as evidence that T inherited this information by deriving from the candidate parent P , we require the additional causal assumption stated below.

Teodora: write this with actual causality

Assumption 1 (Shard-mediated causal model). *The audit is interpreted with respect to a structural causal model over (S, P, T, C) , where S is the secret shard, P is the candidate parent model, T is the target model, and C denotes background factors not specific to the shard.*

The structural assignments have the form

$$P \stackrel{\text{def}}{=} f_P(S, C, U_P), \quad T \stackrel{\text{def}}{=} f_T(P, C, U_T).$$

The associated causal diagram therefore contains the directed path

$$S \longrightarrow P \longrightarrow T,$$

and may contain the background edges $C \longrightarrow P$, and $C \longrightarrow T$. The model satisfies the following restrictions:

1. S is a direct causal parent of P .
2. S is not a direct causal parent of T .
3. **No shard-specific confounding.** *There is no latent factor H that creates dependence between S and T while bypassing P . Formally, the causal model rules out paths of the form*

$$S \longleftarrow H \longrightarrow T.$$

Definition 1 (Causal model provenance). Fix the structural causal model defined in Assumption 1. We say that T is *not* derived from P if intervening on the secret shard does not change the distribution of the target model: for all $s, \tilde{s}$,

$$H_0^{\text{causal}} : T \mid \text{do}(S = s) \stackrel{d}{=} T \mid \text{do}(S = \tilde{s}).$$

We say that T is derived from P if there exists $s, \tilde{s}$ such that

$$H_1^{\text{causal}} : T \mid \text{do}(S = s) \not\stackrel{d}{=} T \mid \text{do}(S = \tilde{s}).$$

Definition 2 formalizes the hypothesis testing problem. The tester observes a tuple (S, S', T) , where S is the secret shard, S' is an independent control shard drawn from the same population, and T is the target model **Ngoc:** does it mean that P does not observe S' ?. The null hypothesis represents the case in which T is unrelated to S , while the alternative represents the case in which T is statistically dependent on S . To model the relationship between T and S , we define a conditional distribution for the target model given the shard. Under the null, the conditional distribution $\Pi_0 \mid S$ does not depend on S , so T is independent of the secret shard. Under the alternative, the conditional distribution $\Pi_1 \mid S$ depends on S , so T may contain shard-specific information. Under Assumption 1, such dependence can be interpreted as evidence that the target model T was derived from the candidate parent P .

Definition 2 (Hypothesis testing problem). For $b \in \{0, 1\}$, let $\Pi_b \mid S$ be a conditional distribution of the target model T given a shard S . The observation under Π_b is a tuple (S, S', T) , which is generated by

$$S, S' \sim \mathcal{D}^n, T \sim \Pi_b \mid S.$$

- Under the *null hypothesis* Π_0 , the conditional distribution $\Pi_0 \mid S$ does not depend on S , thus the target model T is independent of S .
- Under the *alternative hypothesis* Π_1 , the conditional distribution $\Pi_1 \mid S$ depends on S , thus the target model T is statistically dependent on S .

Definition 2 formulates the hypothesis testing problem at the level of joint distributions. In an audit, however, the tester does not know the conditional distribution $\Pi_b \mid S$. Instead, the tester has access to sampled shards S, S' and can query the target model T as a black box. Our key observation is that this dependence-testing problem can be reduced to a distinguishing game, thereby giving an operational test for the hypotheses in Definition 2.

We first describe the distinguishing game at a high level. The game is parameterized by an observation (S, S', T) and a distinguisher A . The distinguisher has access to the construction split $S_{m+1:n}$ and black-box query access to the target model T . It is then evaluated on a single sample x . In the *real* shard experiment, x is sampled from the held-out shard split $S_{:m}$; in the *ideal* shard experiment, x is sampled from the corresponding split $S'_{:m}$ of the independent control shard S' . The goal of the distinguisher is to decide whether its input came from the real or ideal shard experiment. The experiments and the corresponding distinguishing game are formally stated in Definition 3 and Definition 4.

Definition 3 (Real and ideal shard experiments). Let $S, S' \sim \mathcal{D}^n$ independently and $T \sim \Pi \mid S$. Let A be a distinguisher that is given access to $S_{m+1:n}$ and black-box query access to T . We define the real and ideal shard experiments with respect to (S, S', T) as

$$\text{Exp}_{T,S,S'}^{\text{real}}(A) : \begin{array}{l} x \xleftarrow{\$} S_{:m}; \\ \text{return } A(x). \end{array} \quad \text{Exp}_{T,S,S'}^{\text{ideal}}(A) : \begin{array}{l} x \xleftarrow{\$} S'_{:m}; \\ \text{return } A(x). \end{array}$$

Ngoc: should A also have access to $S'_{m+1:n}$? Ngoc: may be $A_T(x)$ or $A(x; T)$

Definition 4 (Shard distinguishing game). Given an audit instance (S, S', T) and a distinguisher A constructed from $(T, S_{m+1:n})$, the shard distinguishing game $\mathcal{D}_{T,S,S'}^{\text{shard}}(A)$ is defined as

$$\begin{array}{l} b \xleftarrow{\$} \{0, 1\}; \\ \text{if } b = 1 \text{ then } \hat{b} \leftarrow \text{Exp}_{T,S,S'}^{\text{real}}(A); \\ \text{if } b = 0 \text{ then } \hat{b} \leftarrow \text{Exp}_{T,S,S'}^{\text{ideal}}(A); \\ \text{return } 1 \text{ iff } \hat{b} = b. \end{array}$$

In the shard distinguishing game (ref Definition 4), the distinguishers are a class of binary-output algorithms. Each distinguisher $A \in \mathcal{A}$ is a (randomized) algorithm with black-box query access to the target model T , and a subset of the shard $S_{m+1:n}$. On input a shard sample $x \in \mathcal{X}$, the distinguisher outputs a bit

$$A(x) \in \{0, 1\},$$

where output 1 means that A predicts that the shard sample x was generated by the real experiment, while output 0 means that A predicts that the shard sample x was generated by the ideal experiment.

We quantify the power of a distinguisher A by its *shard advantage* in the shard distinguishing game. An advantage of 0 means that A cannot distinguish the real experiment from the ideal experiment and performs no better than random guessing. An advantage of 1 means that A perfectly identifies which experiment generated its input. The general case and formal definition is given in Definition 5.

Definition 5 (Shard advantage). For a distinguisher A , define its shard advantage with respect to the observation (S, S', T) by

$$\text{Adv}_A(T, S, S') \stackrel{\text{def}}{=} 2 \left| \Pr \left[\mathcal{D}_{T,S,S'}^{\text{shard}}(A) = 1 \right] - \frac{1}{2} \right|.$$

Equivalently,

$$\text{Adv}_A(T, S, S') = \left| \Pr \left[\text{Exp}_{T,S,S'}^{\text{real}}(A) = 1 \right] - \Pr \left[\text{Exp}_{T,S,S'}^{\text{ideal}}(A) = 1 \right] \right|.$$

To see how the dependence-testing problem in Definition 2 reduces to the shard distinguishing game, observe that the shard advantage $\text{Adv}_A(T, S, S')$ serves as a test statistic. Under the null hypothesis,

160 S and T are independent. Therefore, conditioning on the target model T and the construction split
 161 $S_{m+1:n}$, the held-out shard split $S_{:m}$ and the control split $S'_{:m}$ are two independent samples from
 162 the same population distribution. Consequently, the signed shard advantage in expectation is zero
 163 under the null, and any nonzero value of $\text{Adv}_A(T, S, S')$ arises from finite-sample fluctuation.

164 Proposition 1 formalizes this observation: under the null, every fixed distinguisher A constructed
 165 from $(T, S_{m+1:n})$ has shard advantage at most

$$\sqrt{\frac{\log(2/\alpha)}{m}}$$

166 with probability at least $1 - \alpha$. Thus, a value of $\text{Adv}_A(T, S, S')$ that substantially exceeds this null
 167 fluctuation bound provides evidence that the real shard split is distinguishable from the independent
 168 control split, and therefore supports rejecting the null hypothesis.

169 **Proposition 1** (Shard advantage under the null; Proof is in Section B). *Under the null hypothesis*
 170 Π_0 *of Definition 2, fix any distinguisher A constructed from $(T, S_{m+1:n})$. For every $\alpha \in (0, 1)$, with*
 171 *probability at least $1 - \alpha$ over the draw of (S, S', T) ,*

$$\text{Adv}_A(T, S, S') \leq \sqrt{\frac{\log(2/\alpha)}{m}}.$$

172 Obtaining a good test from the shard distinguishing game then reduces to finding a good distin-
 173 guisher A and estimating its shard advantage $\text{Adv}_A(T, S, S')$ accurately. The distinguisher is con-
 174 structed from the target model T and the construction split $S_{m+1:n}$, while its advantage is estimated
 175 on the held-out shard split $S_{:m}$ and the independent control split $S'_{:m}$. Algorithm 1 defines an em-
 176 pirical estimator for the shard advantage of a fixed distinguisher.

Algorithm 1 Empirical advantage estimator $\text{Est}(A; T, S_{:m}, S'_{:m})$

- 1: **Input:** A distinguisher A , a target model T , the shard split $S_{:m}$, and the control split $S'_{:m}$
 - 2: **Output:** an estimate of $\text{Adv}_A(T, S, S')$
 - 3: Compute $a \leftarrow \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{A(x_i) = 1\}$
 - 4: Compute $a' \leftarrow \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{A(x'_i) = 1\}$
 - 5: Return $|a - a'|$.
-

177 **Proposition 2** (Accuracy of the empirical advantage estimator; Proof is in Section B). *Let (T, S, S')*
 178 *be defined as the above. Then, for every $\alpha \in (0, 1)$, with probability at least $1 - \alpha$ over the internal*
 179 *randomness of the estimator Est ,*

$$|\text{Est}(A; T, S_{:m}, S'_{:m}) - \text{Adv}_A(T, S, S')| \leq \sqrt{\frac{\log(2/\alpha)}{m}}.$$

180 **Lemma 1** (Level- α shard distinguishing test; Proof is in Section B). *Let $\alpha \in (0, 1)$ be a confidence*
 181 *level and $\gamma_\alpha = 2\sqrt{\frac{\log(4/\alpha)}{m}}$. Let A be a fixed distinguisher constructed from $(T, S_{m+1:n})$, and define*
 182 *the thresholding test*

$$\phi_{A,\alpha}(T, S, S') \stackrel{\text{def}}{=} \mathbf{1}\{\text{Est}(A; T, S_{:m}, S'_{:m}) > \gamma_\alpha\}.$$

183 *Then*

184 **Type-I error.** *Under the null hypothesis Π_0 ,*

$$\Pr_{(S, S', T) \sim \Pi_0} [\phi_{A,\alpha}(T, S, S') = 1] \leq \alpha.$$

185 **Instance witness statement.** *When fixing any observation (T, S, S') , if $\text{Adv}_A(T, S, S') >$*
 186 *$\gamma_\alpha + \sqrt{\frac{\log(2/\beta)}{m}}$, then, with probability at least $1 - \beta$ over the internal randomness of Est ,*

$$\phi_{A,\alpha}(T, S, S') = 1.$$

187 **Lemma 2** (Causal no-provenance implies the statistical null). Assume Assumption 1 holds. If the
 188 causal no-provenance null

189 H_0^{causal} in Definition 1 holds, then the induced

190 observational distribution belongs to the statistical null Π_0 . Equivalently,

191 the conditional distribution of T given S does not depend on S .

192 **Theorem 1** (Provenance guarantee of the shard test). Assume Assumption 1 holds. Let $\alpha \in (0, 1)$
 193 and let $\phi_{A,\alpha}$ be the level- α shard distinguishing test from Lemma 1. Then the following statements
 194 hold.

195 **Soundness under no provenance.** With probability at least $1 - \alpha$, the test does not falsely report
 196 that T is derived P if T does not.

197 **Completeness under a detectable provenance signal.** Fix any observation (T, S, S') , if the
 198 derivation leaves a detectable shard signal for the distinguisher A , in the sense that

$$\text{Adv}_A(T, S, S') > \gamma_\alpha + \sqrt{\frac{\log(2/\beta)}{m}},$$

199 then, except with probability at most α , the test correctly report shard-dependent evidence of
 200 provenance when the shard-specific information flowed to T through the candidate parent P .

201 5 Large Language Model Provenance Estimator

202 We instantiate the shard-distinguishing framework of Section 4 for autoregressive language mod-
 203 els. Given black-box access to a target model T , each tokenized document $x = (y_1, \dots, y_r)$ is
 204 represented by its truncated per-token loss trace

$$L_T(x) = (-\log p_T(y_i | y_{<i}))_{i=1}^\ell, \quad \ell = \min(r, L_{\max}),$$

205 with padding applied when needed.

206 Let μ_1^T and μ_0^T denote the distributions of $L_T(x)$ when x is drawn from the real shard split and the
 207 independent control split, respectively. A distinguisher may use the full loss trace, or it may first
 208 compress the trace through an aggregation map g , such as mean loss $[\]$ or a finite set of summary
 209 statistics $[\]$ Ngoc: Min-K?.

210 **Proposition 3** (Optimality of loss traces). The optimal distinguishing advantage achievable from
 211 the full loss trace is

$$\sup_A \text{Adv}_A(T, S, S') = \text{TV}(\mu_1^T, \mu_0^T),$$

212 and is attained by the likelihood-ratio test

$$A_T^*(\ell) = \mathbf{1} \left[\frac{d\mu_1^T}{d\nu}(\ell) \geq \frac{d\mu_0^T}{d\nu}(\ell) \right],$$

213 where ν dominates both μ_1^T and μ_0^T . Furthermore, for any measurable aggregation g ,

$$\text{TV}(g_\# \mu_1^T, g_\# \mu_0^T) \leq \text{TV}(\mu_1^T, \mu_0^T).$$

214 Thus no scalar or finite-dimensional aggregation of the trace can contain more provenance signal
 215 than the full sequence.

216 This proposition motivates sequence-based provenance estimators: rather than testing membership
 217 using only an aggregate loss score, we train a distinguisher on labeled, the full loss traces, and
 218 evaluate its shard advantage on held-out real and control examples.

219 **Sequence-based membership distinguisher.** We instantiate the distinguisher as a one-layer bidi-
 220 rectional GRU [Chung et al., 2014] $\phi_\theta : \mathbb{R}^{L_{\max}} \rightarrow [0, 1]$ with hidden dimension $H = 128$. For a
 221 loss trace $L_T(x)$, the GRU representation is

$$h_{\text{cat}} = [\vec{h}_{L_{\max}}; \overleftarrow{h}_1] \in \mathbb{R}^{2H}.$$

222 A linear layer and sigmoid map this representation to a membership score $\phi_\theta(L_T(x))$. For a thresh-
 223 old τ , the induced binary distinguisher is

$$A_{\theta,\tau}^T(x) = \mathbf{1} \{ \phi_\theta(L_T(x)) \geq \tau \}.$$

224 Output 1 predicts the real shard and output 0 predicts the control shard.

225 Here, the GRU does not add information beyond the black-box loss trace; it only provides a flexible
 226 approximation to the optimal trace distinguisher. In particular, it can capture position-dependent
 227 effects, local runs of low loss, and correlations across adjacent token losses that scalar summaries
 228 discard.

229 6 Experiment Setups

230 6.1 Problem Instantiation

231 We consider the problem of detecting whether a target model T exhibits membership signal with
 232 respect to a candidate shard S drawn from the training distribution of a parent model P . Under
 233 the black-box access model, the auditor observes only per-token log-probabilities from T . **Teodora:**
 234 **Doesn't the auditor know the shard of the parent model too, not clear what the auditor has, is the**
 235 **auditor the parent?**

236 Given two datasets—a member set S and a non-member set S' —our goal is to construct a classifier
 237 that distinguishes whether an input sequence x was drawn from S or S' , based solely on the loss se-
 238 quence induced by T . Under the causal assumption of Section 4, successful discrimination provides
 239 empirical evidence that T retains information about S .

240 We additionally evaluate the same procedure on the parent model P for diagnostic comparison,
 241 although the theoretical framework does not require classification on P for provenance testing.

242 6.2 Data Construction

243 The member dataset S is constructed from Pile-CC, a Common Crawl-derived corpus included
 244 in the pretraining mixture of Pythia models. The non-member dataset S' is drawn from FineWeb,
 245 restricted to documents published after January 2021, ensuring temporal separation from the training
 246 period of the parent model.

247 For the OLMo experiment, we additionally construct an alternative pair of datasets. The member
 248 dataset S is derived from the Paloma Common Crawl subset, which has a cutoff date of March
 249 2023. The corresponding non-member dataset S' is obtained by querying FineWeb for documents
 250 published after April 2023, ensuring strict post-cutoff temporal separation relative to the Paloma
 251 training distribution.

252 Both dataset pairs originate from Common Crawl sources, which reduces coarse domain mismatch
 253 and ensures that separability cannot be attributed to obvious differences in topic distribution or
 254 writing style. This design choice isolates signals arising from model exposure rather than dataset
 255 composition.

256 6.3 Dataset Validation

257 **TODO: Update this section with validation from Paloma's dataset.** We perform a set of diag-
 258 nostics to verify that S and S' are not trivially distinguishable by surface-level statistics or leakage
 259 artifacts.

260 First, we compute TF-IDF cosine similarity within and across datasets and observe a cross-to-within
 261 similarity ratio of 0.9625, indicating that both datasets exhibit highly similar lexical structure con-
 262 sistent with web-scale text corpora.

263 To rule out data leakage, we measure exact string overlap and find zero duplicate documents between
 264 S and S' . Additionally, 7-gram overlap is negligible (0.0157%), confirming the absence of verbatim
 265 reuse across splits.

266 These results suggest that any separability observed in downstream experiments is unlikely to be
 267 driven by direct memorization of shared text across datasets.

268 6.4 Loss Sequence Representation

269 Prior approaches to membership inference typically reduce per-token loss values to a scalar statistic
270 (e.g., mean cross-entropy or Min-K aggregation). While computationally simple, such aggregation
271 discards the structural information present in the loss sequence.

272 We instead represent each document as a sequence of per-token cross-entropy losses:

$$\ell = \{-\log P_T(y_i | y_{<i})\}_{i=1}^L,$$

273 where L is truncated to 1024 tokens.

274 This sequence preserves position-dependent variations in loss, which may reflect localized memo-
275 rization effects. In particular, tokens that are strongly memorized by the model can produce distinc-
276 tive low-loss patterns that are not preserved under scalar aggregation.

277 6.5 Sequence-based Membership Classifier

278 To exploit the structure of the loss sequence, we train a sequence model ϕ that operates directly on
279 ℓ . We use a one-layer bidirectional GRU with hidden dimension $H = 128$.

280 Given an input sequence $\ell \in \mathbb{R}^L$, the GRU produces forward and backward hidden states. The final
281 representation is formed by concatenating the terminal states:

$$h_{\text{cat}} = [\vec{h}_L; \overleftarrow{h}_1] \in \mathbb{R}^{2H}.$$

282 This representation is passed through a linear layer to produce a scalar logit, and membership prob-
283 ability is obtained via a sigmoid activation.

284 A sequence model is necessary in this setting, as per-token losses are not independent and reflect the
285 autoregressive structure of the underlying language model. By modeling temporal dependencies, the
286 classifier can detect structured patterns in loss sequences that distinguish member from non-member
287 data.

288 6.6 Training Procedure

289 The classifier is trained using binary cross-entropy with logits. We use the Adam optimizer with
290 learning rate 10^{-3} , batch size 128, and train for 10 epochs. The dataset is split into 50,000 training
291 examples and 10,000 test examples for each of S and S' .

292 All input sequences are truncated to 1024 tokens. The input dimension is 1 (per-token loss), and the
293 classifier contains approximately 100K parameters.

294 7 Evaluation

295 We ask the following questions:

- 296 1. How effective is our approach at detecting model provenance?
- 297 2. Is our method sensitive to lower shard sizes, X , Y ?

298 7.1 Experimental Setup

299 **Models.** We evaluate across multiple model families. Parent models are drawn from the Pythia
300 suite (1B, 1.4B, 6.9B, 12B), while target models include fine-tuned Pythia checkpoints and
301 OLMo/OLMo2 variants (SFT, instruction-tuned, and DPO).

302 **Datasets.** We construct balanced datasets with 60,000 member (S) and 60,000 non-member (S')
303 records. Each dataset is split into 50,000 training and 10,000 test samples.

304 **Metrics.** We report AUC-ROC as the primary metric.

305 **Baselines.** We use average as a baseline for our approach.

7.2 Main Results (Target Models)

Table 1: Membership inference performance on target models.

Model	Ours (AUC)	Avg Loss	Min-K
<i>Pythia Targets</i>			
Pythia-1.4B (ShareGPT)	0.8847	0.4947	0.4809
Pythia-1.4B (Synthetic)	0.7259	0.5551	0.5469
Pythia-12B (OpenAssistant)	0.8699	N/A	N/A
Pythia-12B (Synthetic)	0.8718	0.6099	0.6126
Pythia-1B (TLDR)	0.9777	0.5478	0.5359
Pythia-6.9B (Alpaca)	0.7696	0.5416	0.5338
Pythia-6.9B (Clinical)	0.8827	0.6019	0.6005
Pythia-6.9B (HC3)	0.9171	0.5302	0.5174
Pythia-6.9B (OpenAssistant)	0.9430	0.5567	0.5689
Pythia-6.9B (TLDR)	0.9767	0.5786	0.5615
<i>OLMo Targets</i>			
OLMo2-7B (DPO)	0.7116	0.4541	0.4984
OLMo2-7B (Instruct)	0.7184	0.4554	0.5005
OLMo2-7B (SFT)	0.7267	0.4449	0.4854
OLMo-7B-0424 (Instruct)	0.7494	0.4325	0.4831
OLMo-7B-0424 (SFT)	0.7505	0.4292	0.4788
OLMo-7B-0724 (Instruct)	0.7451	0.4337	0.4850
OLMo-7B-0724 (SFT)	0.7531	0.4292	0.4802
OLMo-7B (Instruct)	0.7554	0.4674	0.5318
OLMo-7B (SFT)	0.7743	0.4481	0.5030

Across all target models, our sequence-based approach consistently outperforms scalar baselines. Improvements are especially pronounced in Pythia-based models, where AUC reaches up to 0.9777. Even in weaker settings, performance remains substantially above both average-loss and Min-K baselines.

7.3 Parent vs Target Analysis

Table 2: Parent vs target AUC comparison (Pythia models).

Experiment	Parent AUC	Target AUC
Pythia-1.4B (ShareGPT)	0.9429	0.8847
Pythia-1.4B (Synthetic)	0.9462	0.7259
Pythia-12B (OpenAssistant)	0.9406	0.8699
Pythia-12B (Synthetic)	0.9396	0.8718
Pythia-1B (TLDR)	0.9405	0.9777
Pythia-6.9B (Alpaca)	0.9402	0.7696
Pythia-6.9B (Clinical)	0.9423	0.8827
Pythia-6.9B (HC3)	0.9392	0.9171
Pythia-6.9B (OpenAssistant)	0.9425	0.9430
Pythia-6.9B (TLDR)	0.9417	0.9767

TODO: Explain the contradictory AUC scores for parent and target. Specifically 1B TLDR and 6.9B TLDR

7.4 Controls and Validity

We verify that the observed performance is driven by genuine membership signal rather than artifacts.

317 **Control experiment.** In the absence of membership signal, the classifier achieves an AUC of
318 0.5002, which matches random guessing. This confirms that the model does not learn spurious
319 separability when no signal is present.

320 **TODO: Dataset Similarity and extra control experiment**

321 **7.5 Limitations**

322 Our experiments are subject to several limitations.

323 First, each experiment is conducted using a single training run of the classifier. While results are
324 consistent across many model configurations, multiple runs would provide stronger statistical confi-
325 dence and reduce variance.

326 Second, we operate on a subset of available data rather than the full pretraining corpus. A larger
327 and more diverse dataset could improve coverage across domains and potentially strengthen the
328 observed signal.

329 Finally, due to computational constraints, we do not explore the full space of model architectures
330 or hyperparameters. More extensive tuning or alternative sequence models may further improve
331 performance.

332 **8 Discussion and Limitations**

333 We acknowledge the following limitations of the present study.

334 **Single-run variance.** All reported figures reflect a single training run. We have not yet conducted
335 multi-seed experiments; reported AUC values may therefore carry non-trivial variance due to ran-
336 dom weight initialization and batch ordering. We plan to report mean and standard deviation over
337 multiple seeds in the final version.

338 **Generalization of the cuDNN workaround.** The hardware-specific ATen fallback was validated
339 exclusively on the NVIDIA L40S. Reproduction on other GPU architectures (e.g. A100, H100)
340 should re-enable cuDNN and verify numerical equivalence before disabling it.

341 **Dataset scope.** The audit is evaluated on a single member/non-member pairing (Pile-CC vs.
342 FineWeb). Performance may differ on domains with weaker distributional separation or on mod-
343 els fine-tuned on data closer to the non-member distribution.

344 **Black-box assumption.** Our distinguisher requires access to per-token log-probabilities. This is
345 a stronger query interface than pure token-generation access and may not be available from all
346 production API endpoints.

347 **9 Broader Impact**

348 Membership inference audits serve a dual purpose. Defensively, they enable copyright holders and
349 data subjects to verify whether their content was used in model training without consent, supporting
350 emerging data governance and compliance requirements. Offensively, the same techniques could
351 be used to infer the composition of proprietary training sets, which may represent commercially
352 sensitive information. We note that our method requires log-probability access, which is already
353 restricted by most production providers; purely generative APIs are not susceptible to the exact
354 attack described here. We release no new model weights and do not release code at this stage of
355 submission to preserve anonymity; a reproducibility package including seeds, data sampling scripts,
356 and training code will accompany the camera-ready version.

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395 A Detailed Related Work

396 B Missing proofs for Section 4

- 397 *Proof of Proposition 1.* Under the null hypothesis Π_0 , the conditional distribution $\Pi_0 \mid S$ does not
398 depend on S , thereby the target model T is independent of the shard S . Let $(x_1, \dots, x_m) = S_m$

399 By the definition of the real and ideal shard experiments, we then have

$$\begin{aligned}
\Pr \left[\text{Exp}_{T,S,S'}^{\text{real}}(A) = 1 \right] &= \Pr [A(x_I) = 1] & (I \stackrel{\$}{\leftarrow} \{1, \dots, m\}) \\
&= \sum_{i=1}^m \Pr [I = i] \Pr [A(x_i) = 1 \mid I = i] \\
&= \sum_{i=1}^m \frac{1}{m} \Pr [A(x_i) = 1] \\
&= \frac{1}{m} \sum_{i=1}^m \Pr [A(x_i) = 1],
\end{aligned}$$

400 and

$$\Pr \left[\text{Exp}_{T,S,S'}^{\text{ideal}}(A) = 1 \right] = \frac{1}{m} \sum_{i=1}^m \Pr [A(x'_i) = 1].$$

401 Plugging $\Pr \left[\text{Exp}_{T,S,S'}^{\text{real}}(A) = 1 \right]$ and $\Pr \left[\text{Exp}_{T,S,S'}^{\text{ideal}}(A) = 1 \right]$ into the expression of
402 $\text{Adv}_A(T, S, S')$, we have

$$\text{Adv}_A(T, S, S') = \left| \frac{1}{m} \sum_{i=1}^m \Pr [A(x_i) = 1] - \frac{1}{m} \sum_{i=1}^m \Pr [A(x'_i) = 1] \right|.$$

403 Conditional on $(T, S_{m+1:n})$, the samples $x_1, \dots, x_m$ and $x'_1, \dots, x'_m$ are independent
404 draws from $\mathcal{D}$, thereby the random variables $\Pr [A(x_1) = 1], \dots, \Pr [A(x_m) = 1]$ and
405 $\Pr [A(x'_1) = 1] \dots \Pr [A(x'_m) = 1]$ are independent and bounded in $[0, 1]$.

406 Moreover, for every $i, j \in [m]$,

$$\mathbf{E} \mathbf{x} [\Pr [A(x_i) = 1] \mid T, S_{m+1:n}] = \mathbf{E} \mathbf{x} [\Pr [A(x'_j) = 1] \mid T, S_{m+1:n}],$$

407 thus we have

$$\mathbf{E} \mathbf{x} \left[\frac{1}{m} \sum_{i=1}^m \Pr [A(x_i) = 1] - \frac{1}{m} \sum_{i=1}^m \Pr [A(x'_i) = 1] \right] = 0.$$

408 Applying Hoeffding's inequality to the independent bounded summands $\frac{1}{m} \sum_{i=1}^m \Pr [A(x_i) = 1]$
409 and $-\frac{1}{m} \sum_{i=1}^m \Pr [A(x'_i) = 1]$ gives, for every $\tau > 0$,

$$\Pr [\text{Adv}_A(T, S, S') \geq \tau \mid T, S_{m+1:n}] \leq 2 \exp(-m\tau^2).$$

410 Since this bound holds for every realization of $(T, S_{m+1:n})$, it also holds unconditionally:

$$\Pr [\text{Adv}_A(T, S, S') \geq \tau] \leq 2 \exp(-m\tau^2).$$

411 Setting

$$\tau = \sqrt{\frac{\log(2/\alpha)}{m}}$$

412 yields

$$\Pr \left[\text{Adv}_A(T, S, S') \leq \sqrt{\frac{\log(2/\alpha)}{m}} \right] \geq 1 - \alpha,$$

413 which gives the desired statement. $\square$

414 *Proof of Proposition 2.* Let (T, S, S') be the observation and denote by

$$S_{:m} = (x_1, \dots, x_m), \quad S'_{:m} = (x'_1, \dots, x'_m).$$

Let A be the fixed distinguisher constructed from $(T, S_{m+1:n})$. By the definition of the real and ideal shard experiments, we have

$$\Pr \left[\text{Exp}_{T,S,S'}^{\text{real}}(A) = 1 \right] = \frac{1}{m} \sum_{i=1}^m \Pr [A(x_i) = 1],$$

where the probabilities are over the internal randomness of A . Similarly,

$$\Pr \left[\text{Exp}_{T,S,S'}^{\text{ideal}}(A) = 1 \right] = \frac{1}{m} \sum_{i=1}^m \Pr [A(x'_i) = 1].$$

Plugging these two expressions into the definition of $\text{Adv}_A(T, S, S')$, we get

$$\text{Adv}_A(T, S, S') = \left| \frac{1}{m} \sum_{i=1}^m \Pr [A(x_i) = 1] - \frac{1}{m} \sum_{i=1}^m \Pr [A(x'_i) = 1] \right|.$$

On the other hand, the estimator $\text{Est}(A; T, S_{:m}, S'_{:m})$ computes

$$\text{Est}(A; T, S_{:m}, S'_{:m}) = \left| \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{A(x_i) = 1\} - \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{A(x'_i) = 1\} \right|.$$

Define the signed quantities

$$\begin{aligned} \Delta_A(T, S, S') &\stackrel{\text{def}}{=} \frac{1}{m} \sum_{i=1}^m \Pr [A(x_i) = 1] - \frac{1}{m} \sum_{i=1}^m \Pr [A(x'_i) = 1], \\ \widehat{\Delta}_A(T, S, S') &\stackrel{\text{def}}{=} \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{A(x_i) = 1\} - \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{A(x'_i) = 1\}. \end{aligned}$$

Then

$$\text{Adv}_A(T, S, S') = \text{abs} \Delta_A(T, S, S'), \quad \text{Est}(A; T, S_{:m}, S'_{:m}) = \left| \widehat{\Delta}_A(T, S, S') \right|.$$

For every $i \in [m]$, we have

$$\mathbf{E} \mathbf{x} [\mathbf{1}\{A(x_i) = 1\}] = \Pr [A(x_i) = 1], \quad \mathbf{E} \mathbf{x} [\mathbf{1}\{A(x'_i) = 1\}] = \Pr [A(x'_i) = 1].$$

Therefore,

$$\mathbf{E} \mathbf{x} [\widehat{\Delta}_A(T, S, S')] = \Delta_A(T, S, S').$$

Moreover, the random variables $\mathbf{1}\{A(x_1) = 1\}, \dots, \mathbf{1}\{A(x_m) = 1\}$, and $\mathbf{1}\{A(x'_1) = 1\}, \dots, \mathbf{1}\{A(x'_m) = 1\}$ are independent and bounded in $[0, 1]$. Applying Hoeffding's inequality to the independent bounded summands in

$$\widehat{\Delta}_A(T, S, S') = \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{A(x_i) = 1\} - \frac{1}{m} \sum_{i=1}^m \mathbf{1}\{A(x'_i) = 1\}$$

gives, for every $\tau > 0$,

$$\Pr \left[\left| \widehat{\Delta}_A(T, S, S') - \Delta_A(T, S, S') \right| \geq \tau \right] \leq 2 \exp(-m\tau^2).$$

Since the absolute-value map is 1-Lipschitz, we have

$$\begin{aligned} |\text{Est}(A; T, S_{:m}, S'_{:m}) - \text{Adv}_A(T, S, S')| &= \left| \left| \widehat{\Delta}_A(T, S, S') \right| - \left| \Delta_A(T, S, S') \right| \right| \\ &\leq \left| \widehat{\Delta}_A(T, S, S') - \Delta_A(T, S, S') \right|. \end{aligned}$$

Therefore,

$$\Pr [|\text{Est}(A; T, S_{:m}, S'_{:m}) - \text{Adv}_A(T, S, S')| \geq \tau] \leq 2 \exp(-m\tau^2).$$

Setting

$$\tau = \sqrt{\frac{\log(2/\alpha)}{m}}$$

yields the desired statement. $\square$

432 *Proof of Lemma 1.* Let $\eta_\alpha \stackrel{\text{def}}{=} \sqrt{\frac{\log(4/\alpha)}{m}}$, so that $\gamma_\alpha = 2\eta_\alpha$. We first prove the type-I error bound.
 433 Under the null hypothesis Π_0 , applying Proposition 1 with failure probability $\alpha/2$ gives

$$\Pr_{(S,S',T) \sim \Pi_0} [\text{Adv}_A(T, S, S') \leq \eta_\alpha] \geq 1 - \frac{\alpha}{2}.$$

434 Moreover, applying Proposition 2 with failure probability $\alpha/2$ gives

$$\Pr [|\text{Est}(A; T, S_{:m}, S'_{:m}) - \text{Adv}_A(T, S, S')| \leq \eta_\alpha] \geq 1 - \frac{\alpha}{2},$$

435 where the probability is over the internal randomness of the estimator Est .

436 By a union bound, with probability at least $1 - \alpha$ over both the draw of $(S, S', T) \sim \Pi_0$ and the
 437 internal randomness of Est , both events hold simultaneously. On this event, we have

$$\begin{aligned} \text{Est}(A; T, S_{:m}, S'_{:m}) &\leq \text{Adv}_A(T, S, S') + \eta_\alpha \\ &\leq 2\eta_\alpha \\ &= \gamma_\alpha. \end{aligned}$$

438 Therefore,

$$\phi_{A,\alpha}(T, S, S') = 0.$$

439 Equivalently, the event $\phi_{A,\alpha}(T, S, S') = 1$ can occur with probability at most α under the null.
 440 Hence

$$\Pr_{(S,S',T) \sim \Pi_0} [\phi_{A,\alpha}(T, S, S') = 1] \leq \alpha.$$

441 We now prove the instance witness statement. Fix an arbitrary observation (T, S, S') and suppose
 442 that $\text{Adv}_A(T, S, S') > \gamma_\alpha + \sqrt{\frac{\log(2/\beta)}{m}}$. Applying Proposition 2 with failure probability β gives
 443 that, with probability at least $1 - \beta$ over the internal randomness of Est ,

$$|\text{Est}(A; T, S_{:m}, S'_{:m}) - \text{Adv}_A(T, S, S')| \leq \sqrt{\frac{\log(2/\beta)}{m}}.$$

444 On this event,

$$\text{Est}(A; T, S_{:m}, S'_{:m}) \geq \text{Adv}_A(T, S, S') - \sqrt{\frac{\log(2/\beta)}{m}} > \gamma_\alpha.$$

445 By the definition of the thresholding test,

$$\phi_{A,\alpha}(T, S, S') = 1.$$

446 □

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